

AI BACKLOG KILLER

From “Waiting” to “Closed” — redesigning Medical Inquiry operations for Embassy & International Insurance coordination

COO Executive Proposal | Medical Coordination Department | Vejthani Hospital

CURRENT STATE	TARGET STATE	GOVERNANCE PRINCIPLE
20 inquiries currently in backlog Backlog age stated: 3 days Primary blockers: waiting for doctor / verification	≥50% backlog reduction in 30 days Every open case has an owner, next action, due time and escalation point.	AI prepares; humans decide. No AI-only release of medical/financial commitments. Evidence, privacy and approval remain mandatory.

Executive recommendation: treat the backlog as a flow problem—not a people problem. Create one queue, classify every case within minutes, automate reminders and evidence assembly, and reserve human time for medical/financial judgment.

1. Executive Summary

- Diagnosis: the visible backlog is 20 inquiries, but the operational risk is not the count—it is unclear status, dependency on doctors, repeated verification and lack of a hard “next action” clock.
- Recommended operating model: one Backlog Master Queue + standardized intake + AI classification/summarization + automated chase/escalation + human verification + explicit Close-out.
- AI should not be the decision-maker. It should compress reading, extraction, comparison, drafting and follow-up work; staff retain responsibility for medical interpretation, financial commitment, exceptions, privacy and final release.
- 30-day objective: reduce open backlog from 20 to ≤ 10 , while preventing new items from becoming “silent backlog.”
- COO decision required: approve a 30-day pilot, nominate one process owner, agree service levels, and authorize controlled automation for reminders/status tracking.

What success looks like

Metric	Baseline	30-day target	Management signal
Open backlog	20 cases	≤ 10	Red if >10 at Day 30
Aged >3 days	Not yet measured	$<10\%$ of queue	Escalate every breach
Owner assigned	Not standardized	100%	No owner = same-day escalation
Next action + due time	Not standardized	100%	No next action = backlog defect
AI-assisted draft	Not measured	$\geq 60\%$ eligible cases	Track time saved

2. Backlog Diagnosis

Observed fact	Operational interpretation	Risk
20 inquiries are open	Queue size is manageable; flow control is the immediate issue.	Aging compounds if no prioritization exists.
Backlog age stated as 3 days	Cases are already beyond a same/next-business-day response rhythm for many inquiry types.	Embassy/insurer relationship and patient-flow impact.
Waiting for doctor	External dependency is being treated as passive waiting rather than a timed work state.	Decision delay becomes invisible backlog.
Waiting for verification	Evidence assembly and checking are not sufficiently standardized.	Rework, contradictory information, avoidable escalation.

Backlog Data Checklist — required for all 20 cases

Field	Must Have	Why
Case ID / HN / inquiry date	✓	Traceability and aging.
Embassy / insurer / payer	✓	Owner, SLA and communication rule.
Inquiry category	✓	Routing and AI eligibility.
Exact request / decision required	✓	Prevents solving the wrong problem.
Current status + blocker	✓	Separates active work from waiting.
Owner + next action + due time	✓	Creates flow accountability.
Source documents / email thread	✓	Evidence for verification.
Doctor response / financial authority	If applicable	Required for judgment-dependent cases.
Patient sensitivity / PDPA classification	✓	Controls AI/data exposure.

3. Classify — Four-Lane Backlog Model

Lane	Definition	Typical MCO examples	AI leverage	Human judgment
1. AI can do now	Structured, low-risk, evidence-complete tasks.	Extract request; summarize email; create follow-up list; identify missing fields; draft acknowledgement.	High	Low
2. AI prepares; human checks	Draft contains useful work but final response requires verification.	Medical-record summary; insurer response draft; comparison of dates/amounts; embassy status update.	High	Medium
3. Human decision	Decision has clinical, financial, contractual or relationship consequences.	Medical interpretation; coverage exception; commitment of hospital cost; dispute/complaint response.	Medium	High
4. Blocked	Cannot proceed because evidence/authority/system is missing.	Awaiting doctor; missing authorization; conflicting payer instruction; unavailable source.	Low-Medium	High

Critical redesign rule

“Waiting” is not a status that ends the workflow. It must be converted into: Waiting for X → owner Y → requested at time Z → due by T → reminder R → escalation E.

4. Priority Matrix

Score each case 1-5. Higher Impact/Urgency/Risk/AI Leverage increase priority; higher Effort/Dependency reduce immediate priority unless clinically or contractually critical.

Dimension	1	3	5
Impact	Minor / internal	Patient or payer service impact	Embassy/insurer relationship, financial or patient-flow impact
Urgency	>5 business days	Due soon	Due today / patient discharge / payer cutoff
Risk	Low	Moderate	Clinical / financial / contractual / reputational
Effort	<10 min	30-60 min	>2 hours / multiple dependencies
AI Leverage	Mostly manual	AI assists	Highly structured and automatable
Dependency	No dependency	1 dependency	Multiple / external approval

Priority decision rules

- P0 — Immediate: patient-safety, discharge/flight, financial commitment, embassy escalation, or deadline breach risk.
- P1 — Today: complete evidence, low/moderate risk, easy to close; use as Quick Wins to reduce visible queue.
- P2 — Scheduled: complex but important cases; assign a protected work block rather than letting them interrupt the queue.
- P3 — Stop/hold: duplicate, obsolete, no longer requested, or awaiting an authority that has not been re-engaged after defined escalation.

5. Root Cause Analysis

Area	Likely root cause to test	Evidence needed	Countermeasure
Process	No single-state workflow from inquiry → close.	20-case status map; timestamps.	Single Backlog Master Queue + mandatory fields.
People	Doctors are a dependency without a response clock.	Doctor-request timestamps and response time.	Timed medical query + escalation ladder.
Technology	Manual searching, copying and chasing.	Minutes per case by activity.	Email/document extraction + automated reminders.
Data	Information may be fragmented across email/PDF/EMR/Excel.	Source inventory and contradiction log.	Evidence pack and source-of-truth field.
Policy	Approval rules may be implicit.	Who can approve which response.	Approval matrix and exception rules.
Decision delay	Cases remain open while waiting for a decision rather than a next action.	Age by status.	Decision Brief + deadline + escalation.

5 Whys — working hypothesis

Why backlog? → Cases wait. Why? Waiting has no timed owner. Why? The workflow tracks completion, not dependency. Why? Intake/status fields are inconsistent. Why? The process evolved around individual follow-up rather than a queue. Root-cause hypothesis: the system lacks explicit state management and automated escalation. This must be validated against the 20-case timestamps.

Fact vs. assumption: the only confirmed case-level facts provided are 20 open inquiries, 3-day backlog age, and blockers of doctor/verification. The deeper causes above are hypotheses until the queue data is collected.

6. Future-State Workflow — Zero-Based

1. INTAKE	2. TRIAGE	3. EVIDENCE	4. DECISION	5. DRAFT	6. VERIFY	7. CLOSE
Email / inquiry enters one queue. AI extracts sender, payer, patient ref, request and deadline.	AI assigns category, urgency, risk, owner and missing fields.	AI retrieves approved source docs and builds an evidence pack; missing items become tasks.	Doctor / authorized MCO owner decides clinical/financial exception points.	AI generates response using approved template + evidence only.	Human checks accuracy, completeness, source/date, contradiction, compliance and tone.	Send → archive evidence → record decision → timestamp close.

Automation gates

Gate	Automate	Human control
No response	Reminder at T+X; escalation at T+Y.	Owner confirms urgency and escalation recipient.
Missing information	AI identifies missing fields and drafts request.	Staff confirms requested information is appropriate.
Draft response	AI creates evidence-linked draft.	Mandatory human approval before release.
Close	Auto-check required fields and evidence.	Staff confirms outcome and closure reason.

7. AI Opportunities & Tool Mapping

Work step	AI / tool type	Input	Output	Control
Inbox triage	Chat AI + document/email extraction	Email + attachments	Category, urgency, missing data, summary	Check patient/payer identity
Evidence pack	Document AI / search	PDF, Word, approved records	Source list + extracted facts	Source/date/contradiction check
Medical query preparation	Chat AI	Inquiry + relevant facts	1-page Decision Brief for doctor	Doctor owns clinical answer
Draft response	Chat AI / approved template	Decision + evidence + payer context	Embassy/insurer-ready draft	Human approval mandatory
Follow-up	Automation	Status + due time	Reminder / escalation	Escalation rules
Dashboard	Data analysis / BI	Queue timestamps	Aging, SLA, owner, blocker, closure rate	Weekly management review

Technology principle: do not start by buying a new AI platform. Start with a clean queue, standard fields, approved templates and an audit trail; then automate the highest-volume repeatable steps.

8. Generate Draft + Verification System

AI Draft Pattern — no unsupported facts

Section	AI should produce	If data is missing
Acknowledgement	Concise confirmation of request received and current status.	"Additional information required: ..."
Case summary	Patient/payer/request/status in structured form.	Flag "not provided" rather than infer.
Decision requested	One clear question for doctor/authorized approver.	List exact missing decision point.
External response	Draft using approved facts and template.	Do not invent coverage, cost, diagnosis or approval.

10-point Verification Gate

Check	Pass condition	Disposition
Accuracy / completeness	Every material statement traces to evidence.	☐ Verified / ⚠ Review
Source / date	Current approved source identified.	⚠ Review if stale
Context / contradiction	Dates, payer instructions and patient details do not conflict.	☐ Cannot confirm if unresolved
Calculation	Amounts/dates/calculations independently checked.	Human check
Legal / compliance / privacy	Minimum necessary data and policy requirements met.	Human approval
Tone / relationship	Embassy/insurer communication is professional and unambiguous.	Human final read

9. Definition of Done + Close-out

Close criterion	Required evidence	Owner
Request answered / decision completed	Final approved response or documented decision.	Case owner
Medical/financial approval obtained where required	Doctor/authorized approver + timestamp.	Approver
No unresolved contradiction	Verification checklist completed.	Reviewer
External communication completed	Sent email / channel record.	Case owner
Source documents retained	Evidence pack / approved attachments.	Case owner
Next action eliminated	Status = Closed + closure reason.	Case owner

Close-out Checklist

- ☐ Correct patient/payer/inquiry identified
- ☐ Evidence source and date checked
- ☐ Doctor/authorized approval documented when required
- ☐ Final response sent through approved channel
- ☐ Attachments and evidence archived
- ☐ Closure reason recorded
- ☐ Aging and SLA fields updated
- ☐ Any recurrence/root cause captured for weekly review

10. Backlog That Cannot Close — Escalation Model

Blocker	What to do	Owner	Escalate when
A. Waiting for information	AI generates a precise missing-data request; set due time.	Case owner	No response by SLA.
B. Waiting for decision	Create one-question Decision Brief; route to named decision-maker.	Doctor / authorized manager	Decision SLA breached.
C. Waiting for approval	Automated approval task + reminder.	Approver	Second reminder / deadline risk.
D. System limitation	Use controlled fallback; log defect.	IT / process owner	Recurring >2 cases/week.
E. Legal / policy	Stop external release; seek compliance/authorized interpretation.	Compliance / management	Any high-risk uncertainty.
F. Dependency	Track dependency as its own work item, not hidden in the case.	Dependency owner	Dependency breaches SLA.
G. No owner	Assign same day; otherwise manager owns escalation.	MCO manager	Same-day failure.
H. No longer needed	Confirm cancellation/obsolescence; close with reason.	Case owner	No confirmation but aging persists.

Important: “waiting for doctor” should never remain open indefinitely. The correct status is a timed dependency with a next action and escalation owner.

11. KPI — Before / After

KPI	Baseline	30-day target	How measured
Backlog count	20	≤ 10 (–50%)	Open cases at daily cutoff
Average age	3 days stated for current backlog	≤ 2 business days	Close date – open date
Aged >3 days	To measure	<10%	Cases > SLA / total open
Owner assigned	To measure	100%	Queue field completeness
Next action + due time	To measure	100%	Queue field completeness
AI-assisted eligible cases	To measure	$\geq 60\%$	Cases using approved AI step
Average handling time	To measure	–30%	Sampled minutes/case
Rework / correction rate	To measure	–25%	Cases requiring material correction

Do not use AI adoption as the primary KPI. The primary outcome is faster, safer closure; AI utilization is a means, not the goal.

12. Backlog Prevention System

Control	Trigger	Threshold	Owner	Action
Early warning	Case enters queue	No owner within 30 min	MCO shift lead	Auto-assign / escalate
SLA warning	Case age reaches 70% of SLA	Any case	Case owner	Reminder + reprioritize
Escalation	SLA breached	1 breach	Manager	Escalate dependency
Queue health	Daily cutoff	>10 open or >10% aged	Manager	Backlog huddle
Process defect	Same blocker repeats	≥2 cases/week	Process owner	Root-cause fix
Monthly governance	Month-end	Trend deterioration	COO/leadership	Policy/resource decision

Signals that backlog is returning

- More cases are waiting than actively being worked.
- Doctor requests are sent without a due time or escalation owner.
- The same payer/embassy question is repeatedly answered from scratch.
- Staff spend more time searching/retyping than deciding.
- Cases have a status but no next action.
- Closed cases reopen because the response lacked evidence or approval.

13. Action Plan — 7 Days / 30 Days

Timing	Action	Output	Owner
Day 1	Create Backlog Master Queue and load all 20 cases.	20-case baseline with owner/status/age/blocker.	MCO Manager
Day 1-2	Classify every case into 4 lanes + P0-P3.	Prioritized queue.	MCO Lead
Day 2-3	Resolve/close obvious duplicates, obsolete cases and complete-information Quick Wins after confirmation.	First backlog reduction.	Case owners
Day 3-4	Create standard Doctor Decision Brief + timed reminder/escalation.	Faster dependency handling.	MCO + Medical
Day 4-5	Create 3-5 approved response templates and AI prompts.	Controlled draft generation.	MCO Manager
Day 5-7	Pilot AI on low/medium-risk cases with mandatory human verification.	Measured time saved + error log.	Pilot team
Week 2	Automate reminders and queue aging dashboard.	No silent waiting.	MCO + IT
Week 3	Expand AI evidence extraction and draft support to eligible cases.	Higher throughput.	MCO
Week 4	Review KPI, errors, recurrence and resource constraints.	30-day COO review + scale decision.	COO + MCO

30-day success gate

Scale only if: backlog ≤ 10 ; no high-risk AI release without human approval; $\geq 95\%$ queue-field completeness; measurable handling-time reduction; and no material privacy/compliance incident attributable to the pilot.

14. COO Decision Brief — What to Start Today

Three strategic answers

Question	Recommendation
A. Which backlog should not continue?	Any case confirmed as duplicate, obsolete, cancelled, or no longer required. Also stop non-value-added rework such as repeatedly recreating the same status explanation when an approved template/source already exists. Do not close merely because a case is difficult.
B. Where can AI save the most time?	Reading + extraction + comparison + drafting + follow-up. The highest leverage is turning unstructured email/PDF threads into a structured case summary, identifying missing information, preparing a doctor Decision Brief, drafting the external response, and automatically chasing dependencies.
C. Five things to do first for –50% in 30 days	1) Build one 20-case master queue. 2) Triage every case into P0-P3 and four AI lanes. 3) Close confirmed duplicates/obsolete/ready-to-close cases. 4) Put every doctor/approval dependency on a timed escalation clock. 5) Pilot AI drafting + evidence checking on the most repeatable eligible cases.

COO decisions requested

- Approve a 30-day AI Backlog Killer pilot for Medical Coordination.
- Nominate one accountable process owner and one escalation backup.
- Authorize a single queue as the management source of truth.
- Approve controlled AI use with mandatory human verification for external medical/financial communication.
- Require weekly 15-minute backlog review and a Day-30 scale/stop decision.

15. Research Basis, Controls & Limitations

Research basis

- NIST, Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile, 2024 (updated 2026). Used as the risk/governance basis for controlled generative-AI deployment and evaluation.
- WHO, Ethics and governance of artificial intelligence for health, 2021. Used for human autonomy, privacy/confidentiality, accountability, transparency and safety principles.
- Microsoft Learn, Streamline document processing with AI Builder. Used as an example of a practical architecture combining document extraction, automated validation, human review and reporting.
- ISO, The process approach in ISO 9001:2015. Used for process thinking, integrated controls, risk-based thinking and continual improvement.
- U.S. HHS, HIPAA Minimum Necessary guidance. Used as an external benchmark for limiting unnecessary disclosure/access to health information; local Thai PDPA and hospital policy remain controlling requirements.

Limitations / assumptions

- The analysis is based on the information supplied: 20 open inquiries, stated age of 3 days, and blockers of “waiting for doctor / verification.” Individual case data, timestamps, payer type, SLA, value, risk and deadlines were not supplied; therefore case-level scoring is intentionally not fabricated.
- The 30-day targets are proposed management targets, not historical performance claims.
- AI should not independently make clinical decisions, coverage determinations, financial commitments, or final external communications.
- PDPA, contractual payer rules, hospital policy and applicable law must be checked before production deployment.
- The recommended pilot should begin with low/medium-risk administrative inquiry tasks and expand only after measured accuracy and safety.

Bottom line: The fastest route to a 50% reduction is not “more AI.” It is queue discipline + dependency clocks + elimination of non-value work + AI-assisted preparation + human-controlled release.